

Principles of Economics

Week 3: Strategy, Altruism and Cooperation

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Social interaction

Game theory

Nash equilibrium

Social dilemmas and the prisoner's dilemma

Pareto efficiency

Sequential (dynamic) games

Social dilemmas and altruism

1. Social interaction

- ▶ Last week: how individuals choose their hours of work, trading off consumption and free time. [Micro, Unit 3]
- ▶ **Self-interest can produce socially beneficial outcomes:** innovation, entrepreneurship. [Micro, Unit 1]
- ▶ However, **self-interest can be harmful to society:**
 - ▶ Why do these problems arise?
 - ▶ What can we do about them?
- ▶ Today: game theory to model social interactions and explain social dilemmas. [Micro, Unit 4]

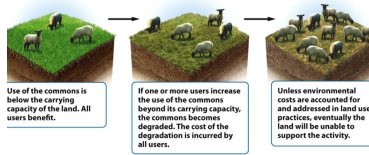
- Social interactions can be good, but also **problematic**:

Belling the cat and self-sacrifice.



The “tragedy of the commons” (Garrett Hardin): overexploitation of common resources, from pastures and fish to air and the climate.

The Tragedy of the Commons



Social dilemma: what is best for one may be worse for society; some free-ride on the efforts of others.

2. Game theory

► John von Neumann



- ▶ **Strategic Interaction:** A situation in which the optimal decision depends on what others do (and vice versa)
- ▶ **Game:** model of a strategic interaction: players, feasible strategies, information and payoffs.
 - ▶ **Player:** A participant in the game.
 - ▶ **Strategy:** a detailed plan of action (and of reaction, in dynamic games).
 - ▶ **Payoffs:** the player's final utility, given the strategies of everyone in the game.
- ▶ **Equilibrium:** the predicted solution of the game (one, many or none).

First game

Set-up

- ▶ Two farmers, Anil and Bala [players], choose which crop to produce.
- ▶ Two options:
 - ▶ Rice or Cassava;
 - ▶ The payoff of each option:
 - ▶ is different for Anil and Bala;
 - ▶ and depends on the decisions of both.
- ▶ We also assume:
 - ▶ Anil and Bala decide simultaneously;
 - ▶ The game is played only once [no repetition];
 - ▶ And they cannot communicate with each other beforehand [no coordination];
 - ▶ The payoffs are common knowledge to both players.



► **Example 1:**

		Bala	
		Rice	Cassava
Anil	Rice	(1, 3)	(2, 2)
	Cassava	(4, 4)	(3, 1)

- Bala gets a payoff of 2 when he chooses cassava and Anil rice.
- Bala's payoff depends on his own decision and on Anil's, which he cannot control.

Solving the game: dominant strategies vs. dominated strategies

- ▶ To solve the game, think about the **players' best responses**:
 - ▶ **Best response**: the strategy with the highest payoff, given what the others do.
 - ▶ Anil's best responses: compare his payoffs across all of Bala's possible actions.
- ▶ Some games are solved by spotting **special strategies**:
 - ▶ **Dominant strategy**: a strategy that is **the best response** regardless of what the others do.
 - ▶ **Dominated strategy**: a strategy that is **NOT the best response** regardless of what the others do.
- ▶ What can we learn from these strategies?
 - ▶ A unique dominant strategy is **always chosen**;
 - ▶ A dominated strategy is **never chosen**.

- ▶ **Example 1:** Two farmers, **Anil** and **Bala**, must decide which crop to plant for the next harvest: **rice** or **cassava**.
- ▶ Because of the land each one has, **Anil** would rather grow **cassava** and **Bala** would rather grow **rice**.
- ▶ The catch: if they both plant the **same** crop, they **flood the market** and the price collapses.
- ▶ So what is best for each one depends on **what the other does**: a **strategic interaction**.

- ▶ **Example 1:** With these payoffs, what is the **outcome of the game**?

		Bala	
		Rice	Cassava
Anil	Rice	(1, 3)	(2, 2)
	Cassava	(4, 4)	(3, 1)

- ▶ What are the **best responses**?
- ▶ Let's start with Bala:
 - ▶ If Anil chooses rice, what does Bala do?
 - ▶ Rice gives him 3, cassava 2: best response, rice.
 - ▶ And if Anil chooses cassava?
 - ▶ Rice gives him 4, cassava 1: best response, rice again.
- ▶ Rice is a **dominant strategy** for Bala.

First game

Solving the game

- **Example 1:** With these payoffs, what is the **outcome of the game**?

		Bala	
		Rice	Cassava
Anil	Rice	(1, 3)	(2, 2)
	Cassava	(4, 4)	(3, 1)

- What are the **best responses**?
- Let's continue with Anil:
 - If Bala chooses rice, what does Anil do?
 - Cassava gives him 4, rice 1: best response, cassava.
 - And if Bala chooses cassava?
 - Cassava gives him 3, rice 2: best response, cassava again.
 - Cassava is a **dominant strategy** for Anil.

First game

Solving the game

- ▶ **Example 1:** With these payoffs, what is the **outcome of the game**?

		Bala	
		Rice	Cassava
Anil	Rice	(1, 3)	(2, 2)
	Cassava	(4, 4)	(3, 1)

- ▶ What is the **equilibrium of the game**?
 - ▶ Where the best responses meet: Bala rice, Anil cassava. Payoffs (4,4)
- ▶ Would anyone change strategy?
 - ▶ No: both are doing their best.

Can we introduce more options?

Dominant Strategies vs. Dominated Strategies

- ▶ More complex strategic situations.
- ▶ **Example 2:** Consider the following game.

		Bala		
		Rice	Cassava	Potato
Anil	Rice	(2, 2)	(4, 1)	(4, 0)
	Cassava	(1, 4)	(0, 0)	(1, 0)
	Potato	(0, 4)	(0, 1)	(0, 0)

- ▶ Find the dominant and dominated strategies.
- ▶ What will the equilibria be?

Can we introduce more options?

Iterated elimination of dominated strategies

- **Example 3:** Consider the following game.

		Bala		
		Rice	Cassava	Potato
Anil	Rice	(2, 2)	(4, 3)	(4, 0)
	Cassava	(1, 4)	(0, 1)	(1, 5)
	Potato	(0, 4)	(0, 1)	(0, 0)

- Find the dominant and dominated strategies.
- What will the equilibria be?

Can we introduce more options?

Iterated elimination of dominated strategies

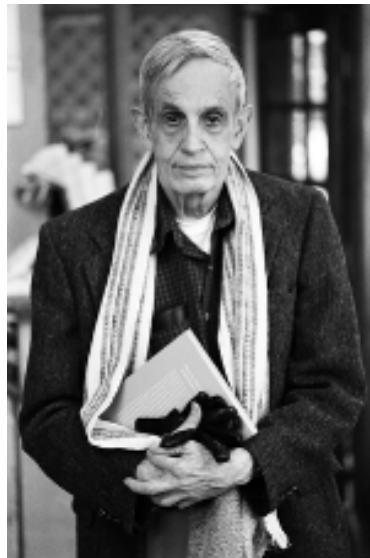
- **Example 4:** Consider the following game.

		Bala		
		Rice	Cassava	Potato
Anil	Rice	(2, 2)	(0, 3)	(0, 0)
	Cassava	(1, 4)	(4, 1)	(1, 5)
	Potato	(0, 4)	(0, 1)	(4, 0)

- Find the dominant and dominated strategies.
- What will the equilibria be?

3. Nash equilibrium

► John F. Nash



- ▶ **Nash equilibrium:** one strategy per player, each a best response to the others. No one gains by deviating unilaterally.
 - ▶ **Example 4:** Recall the game.

		Bala		
		Rice	Cassava	Potato
Anil	Rice	(2, 2)	(0, 3)	(0, 0)
	Cassava	(1, 4)	(4, 1)	(1, 5)
	Potato	(0, 4)	(0, 1)	(4, 0)

- ▶ Every game has a Nash equilibrium (we will not see why in this course)...
- ▶ So we can always solve a game and make a prediction.
- ▶ Food for thought: what is the relation between Nash equilibrium and equilibrium in dominant strategies.

- ▶ **Nash equilibrium:** one strategy per player, each a best response to the others. No one gains by deviating unilaterally.
- ▶ **Example 5:** Consider the following game.

		Bala	
		Rice	Cassava
Anil	Rice	(4, 2)	(1, 1)
	Cassava	(1, 1)	(2, 3)

- ▶ All games have a Nash equilibrium...
- ▶ ...but some have more than one, and then the prediction is not sharp.

- ▶ **Example 6:** The initial game, with different payoffs. What is the **outcome** now?

		Bala	
		Rice	Cassava
Anil	Rice	(0, 1)	(2, 2)
	Cassava	(4, 4)	(1, 0)

- ▶ Solving as before, **the equilibria** are:
 - ▶ Anil produces rice and Bala Cassava, payoffs $\rightarrow (2, 2)$
 - ▶ Anil produces Cassava and Bala rice, payoffs $\rightarrow (4, 4)$
- ▶ With **two Nash equilibria**, does it matter which one happens?
 - ▶ Yes: with (4,4) everyone is better off.
- ▶ How can we induce (4,4)?
 - ▶ Encourage coordination
 - ▶ Governments/institutions intervene to ease coordination in the good equilibrium (week 10)

4. Social dilemmas and the prisoner's dilemma

The prisoners' dilemma

An example of social dilemma

- ▶ Now the **pest-control game**.
 - ▶ Anil and Bala choose which pest-control product to use.
- ▶ There are two options:
 - ▶ **IPC** is less effective, but more environmentally friendly;
 - ▶ **Terminator** is more effective, but far more damaging.
- ▶ Recall we assume:
 - ▶ Anil and Bala **decide simultaneously**;
 - ▶ The game **is played only once** [no repetition];
 - ▶ And they cannot communicate with each other beforehand [no coordination];
 - ▶ **The payoffs are common knowledge** to both players.

The prisoners' dilemma

The social impact of players' choices

► **Example 7:** With these payoffs, what is the **outcome of the game**?

► Are there **dominant strategies**?

► Yes: Terminator is the best response for both.

► What is the **equilibrium of the game**?

► Both use Terminator, payoffs (2,2). Is that the best outcome?

► **Prisoners' dilemma:** a dominant strategy equilibrium whose payoffs, individual and total, are lower than under other strategies.

► It is an example of social dilemma: the optimal individual choice, the equilibrium, gives rise to the worst social outcome.

		Bala	
		IPC	Terminator
Anil	IPC	(3, 3)	(1, 4)
	Terminator	(4, 1)	(2, 2)

Prisoners' Dilemma

The social impact of players' choices

Why do we call it the **Prisoners' Dilemma**?

Another formulation of the game: "Two members of a criminal gang are arrested. Each prisoner is in solitary confinement with no possibility of speaking to or exchanging messages with the other. The police admit they do not have enough evidence to convict the pair on the principal charge. They plan to sentence both to a year in prison on a lesser charge. At the same time, the police offer each prisoner a bargain with the devil". More on [wikipedia](#).

- ▶ **Example 8:** Thelma and Louise are arrested and interrogated separately. Each can accuse the other or deny. Payoffs on the right.
- ▶ Find the dominant strategies and the equilibrium.

Years in prison

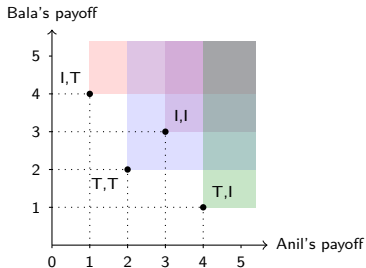
		Louise	
		Deny	Accuse
Thelma	Deny	(1, 1)	(10, 0)
	Accuse	(0, 10)	(5, 5)

- ▶ The prisoners' dilemma is an example of a **social dilemma**: what is best for **society** conflicts with what is best for the **individual**.
- ▶ Game theory shows that in these cases we can end up in an **unambiguously regrettable** outcome.
- ▶ How is it solved **in practice**?
 - ▶ **Reputation** → modelled with *repeated games*.
 - ▶ **Institutions and policy** → modelled as *changing the rules of the game* (mechanism design).
 - ▶ **Spontaneously, altruism** solves it → *the theory of social preferences*: we shall study at the end.

5. Pareto efficiency

Efficiency

- ▶ We need **criteria to judge the solutions**. One of them: **efficiency**.
- ▶ **Pareto efficiency**: no one can be made better off without making someone else worse off.
- ▶ **Alternative definition**:
 - ▶ An outcome is **Pareto dominated** if there exists another outcome in which nobody is worse off but at least one agent is strictly better.
 - ▶ An outcome is Pareto efficient if it is not Pareto dominated.



6. Sequential (dynamic) games

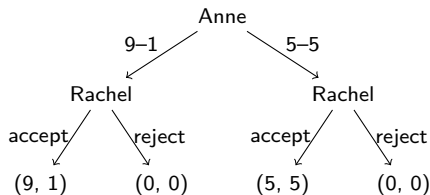
Sequential (dynamic) games

What are they?

- ▶ So far, **static** games: players choose **simultaneously** (without seeing the other's choice).
- ▶ In a **sequential** (dynamic) game players choose **in turns**: one decides and the other **observes and reacts**.
 - ▶ **The order** of moves and **who knows what** at each point matter.
- ▶ New tools: the **game tree** and **backward induction**.
- ▶ Guiding example: **the ultimatum game** (dividing a “pie”).
 - ▶ **The proposer** (Anne) offers a split; **the responder** (Rachel) accepts or rejects.
 - ▶ If she rejects, **no one gets paid**.

Sequential (dynamic) games

The game tree and backward induction



Payoffs: (Anne, Rachel). Backward-induction prediction:

(9, 1).

- ▶ The **tree** represents the game: **nodes** (whose turn it is), **branches** (actions) and **payoffs** at the end.
- ▶ **Backward induction**: solve **from the end**.
 - ▶ A selfish Rachel **accepts any amount** > 0 (something beats nothing).
 - ▶ Anticipating this, Anne keeps **almost everything** (a 9-1 split).
- ▶ Solution: **subgame-perfect equilibrium** = (9, 1).

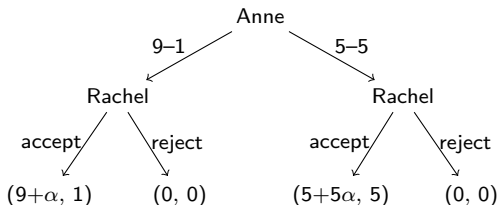
7. Social dilemmas and altruism

- ▶ **Game theory** is the **framework**: players, strategies, information, payoffs and equilibrium.
- ▶ **Preferences** are another **dimension**: **what agents care about** and how they **evaluate** outcomes.
- ▶ So far we assumed them **selfish**: they value only their **own payoff**.
- ▶ But an important dimension of social interaction is **social preferences**: sometimes agents **care about others** (they are not selfish).
- ▶ Here we explore the consequence of introducing **altruistic** preferences into strategic interaction.

- ▶ It helps to separate two things:
 - ▶ **Material payoff** (m): what each one **receives** (money, outcome).
 - ▶ **Evaluation** (utility u): **how they value it**.
- ▶ **Selfish**: $u_i = m_i$ (only their own payoff counts).
- ▶ **Altruistic**: also values the other's payoff, $u_i = m_i + \alpha m_j$, with $\alpha \geq 0$ (how much they care about the other).
- ▶ Going from selfish to altruistic means **replacing the payoffs with their evaluation** and **re-solving the same game**.

Social preferences

The ultimatum game with an altruistic proposer



Payoffs: (Anne's evaluation $u_A = m_A + \alpha m_R$, Rachel's payoff m_R).

- ▶ **The same tree** as in the selfish case, but Anne's payoff is now her **evaluation** $u_A = m_A + \alpha m_R$.
- ▶ **Backward induction:** Rachel accepts (her payoff > 0).
- ▶ A **selfish** Anne ($\alpha = 0$) $\rightarrow$ **9-1**.
Question: from what value of α does Anne choose **5-5**?
- ▶ Valuing the other's payoff **enough changes the equilibrium:** from $(9, 1)$ to $(5, 5)$. Social preferences can **ease the dilemma**.

Section by section (Micro, Unit 4)

1. **Social interaction** → self-interest can help society, and can also harm it: the social dilemma.
2. **Game theory** → players, strategies, information, payoffs; solved with best responses and dominant/dominated strategies.
3. **Nash equilibrium** → every game has one, some have several, and then the prediction is not sharp.
4. **Social dilemmas and the prisoner's dilemma** → everyone can be worse off in equilibrium: an unambiguously regrettable outcome.
5. **Pareto efficiency** → a criterion to judge outcomes: the equilibrium need not be efficient.
6. **Sequential (dynamic) games** → game tree and backward induction; the subgame-perfect equilibrium, illustrated with the ultimatum game.
7. **Social dilemmas and altruism** → social preferences change how payoffs are evaluated; valuing the other enough changes the outcome.

Key concepts: strategic interaction, game, strategy, payoff, best response, dominant and dominated strategy, Nash equilibrium, prisoners' dilemma, Pareto efficiency, sequential (dynamic) game, game tree, backward induction, subgame-perfect equilibrium, ultimatum game, social preferences, altruism.